Mike Ichikawa

DATA SCIENTIST · ML ENGINEER · DATA ENGINEER

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SUMMARY

AI/ML and agent-orchestration engineer with a strong quantitative core who builds and ships production systems and reasons carefully about what they output. Solo and end to end, I designed and shipped a production AI agent for Microsoft 365 (Founding AI/ML Engineer, Microclaw LLC), now live on the Microsoft Commercial Marketplace, with KNN tool selection, RAG semantic memory, smart model routing, and an in-house eval framework. Before engineering, nine years teaching undergraduate mathematics at Santa Rosa Junior College and three years of semiconductor physical design at Intel. MS Mathematics (CCNY, 2014), BS Mechanical Engineering (UC Berkeley). Public portfolio spans time series forecasting, anomaly detection, NLP, cloud ETL, ML serving, a Databricks medallion lakehouse, a production-shape GCP RAG pipeline (Vertex AI Gemini + BigQuery Vector Search + LangChain/LlamaIndex), C++ pybind11 extensions, and a five-repo paper trading system.

TECHNICAL SKILLS

Languages	Python · SQL · C++ · R · TypeScript · Node.js
ML / Statistics	scikit-learn · Prophet · PyTorch · XGBoost · LightGBM · scipy · statsmodels · Bayesian inference
LLM / NLP	Anthropic API · Azure OpenAI · Microsoft Graph · MSAL · HuggingFace · FinBERT · prompt engineering
Data Engineering	PostgreSQL · Databricks · Delta Lake · medallion · dbt · SQLAlchemy · pandas
Cloud / Infra	AWS (S3 · Lambda · DynamoDB · CloudWatch) · Docker · FastAPI · Redis · Pydantic v2 · boto3 · Microsoft Commercial Marketplace
Tools	Git · pytest · Vitest · Jupyter · Streamlit · Docker Compose

EXPERIENCE

Founding AI/ML Engineer
2026 – present

Microclaw LLC · Portland, OR

- Built the agent orchestration harness end to end: an iterative plan/act/observe loop (Azure OpenAI GPT-4o / GPT-4o-mini, function-calling) that chains up to 35 tool calls per task across 84 Microsoft Graph tool functions and 12 Microsoft 365 services, with multi-layer failure recovery (backoff + retry across the LLM and Graph APIs, loop detection); live on Microsoft AppSource (Standard + Self-Hosted offerings)
- Implemented KNN tool selection over 1,045 synthetic training examples using `text-embedding-3-small` (k=7, 0.80 confidence threshold, cosine-similarity fallback): 95.0% recall vs 92.7% embedding-only baseline, selecting ~6 of 84 tools per request vs 18
- Designed RAG-pattern semantic memory retrieval (cosine similarity over per-user persistent context) and a smart model router (GPT-4o-mini default, mid-conversation escalation to GPT-4o on tool-call complexity; the mini path is ~24× cheaper per call); built an in-house eval framework scoring agent output across 107 tests against a baseline bot
- Built a natural-language automations layer where users describe IFTTT-style rules in plain English; the agent parses intent into structured rules persisted to Azure Postgres, fired in real-time via Microsoft Graph change-notification webhooks. 374 Vitest unit tests; Postgres row-level security for tenant isolation; Bicep IaC; Docker → GHCR release pipeline via GitHub Actions

Mathematics Faculty
2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for nine years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Designed assessments and iterated course materials from cohort performance data
- Taught STEM majors alongside non-STEM majors fulfilling general-education math requirements at an open-enrollment community college

Physical Design Engineer
2008 – 2011

Intel Corporation · Semiconductor R&D

- Microprocessor physical design on Intel Ivy Bridge (22nm), including timing closure, power analysis, and design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run

EDUCATION

MS Mathematics — The City College of New York — 2014

BS Mechanical Engineering — University of California, Berkeley — 2008

PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Trading System Arc
Live Demo · github.com/mtichikawa — 5 repos

Five interconnected repositories: live OHLCV pipeline (ccxt/Kraken + PostgreSQL) → candlestick chart generation (mplfinance) → dual-path signal engine (EMA/RSI/MACD/BB + local FinBERT sentiment, 0.6/0.4 fusion) → parameter-optimizing backtester (Sharpe/Sortino/drawdown, staged grid search, T3 feedback loop) → Streamlit multi-page oversight dashboard with live signal display, equity curve, and parameter review. 174/174 tests across T2–T5.

ccxt · PostgreSQL · mplfinance · FinBERT · pandas · NumPy · Streamlit · Plotly · pytest

Databricks Medallion Lakehouse
github.com/mtichikawa/databricks-lakehouse

Medallion lakehouse (bronze → silver → gold) on 10M NYC taxi rows with Delta Lake. Append-only bronze with provenance metadata; silver types, dedupes, and quarantines invalid rows rather than dropping them; gold serves query-specific aggregates. Row-level data quality gates between layers halt the pipeline on schema drift or null-rate breaches. 95 tests across bronze/silver/gold/quality/pipeline. Runs locally via deltalake; deployable to Databricks Community Edition.

Delta Lake · medallion architecture · pandas · pyarrow · data quality gates · pytest

Real-Time Anomaly Detection
github.com/mtichikawa/anomaly-detection

Streaming ensemble detector: Isolation Forest, LSTM autoencoder, and Z-score with 2-of-3 majority voting. Dockerized FastAPI REST API with health checks and configurable detector selection. Live interactive demo on portfolio site.

Docker · FastAPI · scikit-learn · PyTorch · Redis · pytest

Cloud ETL Pipeline
github.com/mtichikawa/cloud-etl-pipeline

AWS-native ETL: tenacity-based REST API ingestion with exponential backoff → Parquet/Snappy storage on S3 with Hive date partitioning → Lambda validation → DynamoDB run logging for failure traceability. Tests mock all AWS calls with moto.

AWS S3 · Lambda · DynamoDB · boto3 · pandas · Apache Parquet · moto

SQL Analytics Pipeline
github.com/mtichikawa/sql-analytics-pipeline

Three-layer dbt-style pipeline (staging → clean → marts) on NYC Airbnb data. Window functions, CTEs, SQLAlchemy ORM with swappable SQLite/PostgreSQL backend. Layered transform pattern enforces raw data immutability.

PostgreSQL · SQLAlchemy · dbt-style transforms · SQLite · pandas

LLM Data Analysis Assistant
github.com/mtichikawa/llm-data-assistant

Two-path query routing: rule-based fast path handles ~70% of queries (correlations, aggregations, group-bys) with zero API cost. Queries below confidence threshold escalate to Anthropic API with DataFrame context injection and multi-turn conversation history.

Anthropic API · pandas · hybrid routing · multi-turn conversation

Additional: Multi-Armed Bandit A/B Testing (Thompson Sampling · UCB1 · Bayesian inference) · Bias Detection in LLMs (ANOVA · Cohen's d) · Financial NLP Parser (SEC EDGAR · regex · sentiment lexicon) · GitHub Trend Forecaster (Prophet · PyTorch LSTM in progress)